

DSA

1) List of companies conducting placement drives using DSA?

<u>Tier-1</u>	<u>Tier 2</u>	<u>Indian</u>
Google	Oracle	Flipkart
Microsoft	Intel	Phone Pe
Amazon	Qualcomm	Swiggy
Apple	Samsung	Zomato
Adobe	Paypal	Meesho
NVIDIA	Goldman Sachs	Groww
Uber	Morgan Stanley	CRED
LinkedIn		Zoho
Meta		Freshworks

Service - Based

Tata	Accenture
Infosys	Capgemini
Cognizant	HCL Tech
Wipro	LTIMindtree

2) What is the syllabus of DSA in Gate?

1. Programming & Data Structures

- Programming in C
- Functions
- Recursion
- Arrays
- Strings

- Linked Lists [singly, Doubly, Circular]
- stacks
- Queues [simple, Circular, Priority]
- Trees

(i) Binary Trees

(ii) Binary Search Trees (BST)

→ Binary Heaps

→ Graphs

Algorithms

→ Asymptotic Notations

1) Big-O (O)

2) Omega (Ω)

3) Theta (Θ)

→ Time Complexity Analysis

→ Space Complexity Analysis

→ Searching Algorithms

1) Linear Search

2) Binary Search

→ Sorting Algorithms

1) Bubble Sort

2) Selection Sort

3) Insertion Sort

4) Merge Sort

5) Quick Sort

6) Heap Sort

→ Hashing

→ Algorithm Design Techniques

1) Divide and Conquer

2) Greedy Algorithms

3) Dynamic Programming

→ Graph Algorithms

1) Breadth-First Search [BFS]

2) Depth-First Search [DFS]

3) Minimum Spanning Tree
[Prim's and Kruskal's]

4) Shortest Path Algorithms

[Dijkstra, Bellman-Ford, etc.]

Q) Find out what are the set of variables in Amazon invoice bill.

A

Order Information

1) Order ID

2) Order Date

3) Invoice Number

4) Invoice Date

5) Invoice Details Reference

Seller Information

1) Seller Name

2) Seller Address

3) GSTIN

4) PAN Number

5) CIN Number

Buyer Information

- 1) Customer Name
- 2) Billing Address
- 3) Shipping Address
- 4) State Code
- 5) Place of Supply
- 6) Place of Delivery

Product Information

- 1) Product Name
- 2) Product Description
- 3) ASIN
- 4) Quantity
- 5) Unit price
- 6) Total price per Item

Tax Information

- 1) Taxable Value
- 2) GST
- 3) CGST
- 4) SGST
- 5) IGST
- 6) Tax Total

Invoice Totals

- 1) Subtotal
- 2) Shipping Charges
- 3) Discount
- 4) Grand total

Logistics Information

- 1) Shipping Address
- 2) Delivery Date
- 3) Shipment ID

- 4) Find out a set of variables and datatypes in Gas Bill, Current Bill,

Gas Bill

variable	Data Type
Bill - ID	String
customer - ID	string
Customer - Name	string
Address	string
Provider - Name	string
Meter - Number	string
Billing - Period	Date Range
Bill - Date	Date
Due - Date	Date
Current - Meter - Reading	Float
Gas - Consumed	Float
Unit - Rate	Decimal
Fixed - Charges	Decimal
Gas - Charges	Decimal

Tax

Decimal

Discount

Decimal

Late-Fee

Decimal

Total-Amount

Decimal

Amount - Paid

Decimal

Payment - Status

String

Payment - Method

String

Payment - Date

Date

Bill - ID
Customer - ID
Customer - Name
Address
Provider - Name
Member - Number
Billing - Period
Bill - Date
Due - Date
Current - Meter - Reading
Gas - Consumed
Unit - Rate
Fixed - Charge
Gas - Charge
Date Range
Date
Date
Float
Float
Decimal
Decimal
Decimal